

Member ID: _____

Time: _____

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SQL DATABASE FUNDAMENTALS (345) **STATE 2023**

Total Point (50 @ 2 points each) _____(100 points)

Test Time: 60 minutes

GENERAL GUIDELINES:

Failure to adhere to any of the following rules will result in disqualification:

1. Member must hand in this test booklet and all printouts if any. Failure to do so will result in disqualification.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.

Directions: Identify the letter of the choice that *best* completes the statement or answers the question.

1. What output would the following query produce – `SELECT "Res""ult"?`
 - A. Result
 - B. Res"ult
 - C. Res
 - D. No output
2. What output would the following query produce – `SELECT 8*8 FROM Customers?`
 - A. No result
 - B. 8 repeated 8 times
 - C. 64
 - D. 64 repeated for every record in the customer table
3. What output would the following query produce – `SELECT * FROM Customers ORDER BY field1 ASC, field1 DESC?`
 - A. All columns in the Customers table order by field1 descending (Z-A)
 - B. All columns in the Customers table order by field1 ascending (A-Z)
 - C. All columns in the Customers table with no order applied
 - D. No result
4. Which of the following is NOT a DBMS?
 - A. PostgreSQL
 - B. Access
 - C. MySQL
 - D. VeriSQL
5. LIMIT is the ANSI standard way to restrict the number of rows returned in a query?
 - A. True
 - B. False
6. “`SELECT field1, MAX(field2) FROM Customers`” will return _____.
 - A. the maximum value in field2 and the field1 that corresponds to it
 - B. the maximum value in field2 and a potentially unrelated field1
 - C. the maximum value in field2 and the last field1 in the table
 - D. the maximum value in field2 and the first field1 in the table

7. To gather all of rows where field1 is either 'A', 'B', or 'C' the WHERE clause would be "WHERE _____".
- A. field1 = ('A', 'B', 'C')
 - B. field1 = 'A' or 'B' or 'C'
 - C. field1 = 'A' and field1 = 'B' and field1 = 'C'
 - D. field1 in ('A', 'B', 'C')
8. _____ are all DML statements.
- A. INSERT, UPDATE, and DELETE
 - B. SELECT, WHERE, ORDER BY, and INTO
 - C. AS, ALIAS, and CONCAT
 - D. Transaction, Roll Back, Backup
9. The UNION operator is used to _____.
- A. combine the result sets of two or more SELECT statements
 - B. uniquely identify the rows in a table
 - C. combine two disparate tables into one
 - D. extract the columns from one table to alter the columns from a second table
10. "_____" would add a new field to the Customers table with the name email.
- A. MODIFY TABLE Customers ADD COLUMN Email varchar(255)
 - B. ALTER TABLE Customers NEW COLUMN Email varchar(255)
 - C. ALTER TABLE Customers ADD Email varchar(255)
 - D. MODIFY TABLE Customers NEW Email varchar(255)
11. Prepared SQL code that can be stored and called over and over is call a stored _____.
- A. object
 - B. function
 - C. procedure
 - D. method
12. Which of the following is NOT an SQL logical operator?
- A. LESS
 - B. SOME
 - C. ALL
 - D. ANY

13. Given the table mytable, where field1 is either 1 or 0 and field2 is either 1 or 0, what will be the result of the query “SELECT field1|field2 FROM mytable”?
- A. All rows from mytable with field1 and field2 concatenated together
 - B. Only rows from mytable where either field1 or field2 is 1
 - C. Field1 if its 1, else field2 if its 1, otherwise no row
 - D. A 0 for each row where field1 and field2 are 0; all other rows will be 1
14. A GROUP BY clause cannot contain aggregate functions?
- A. True
 - B. False
15. _____ are used to restrict the type of data that can go into a table.
- A. Simple Keys
 - B. Views
 - C. Constraints
 - D. Wildcards
16. What of the following is NOT a true statement about the differences between primary and candidate keys?
- A. A primary key cannot be null, but a candidate key can
 - B. A primary key is a candidate key
 - C. A table can have multiple primary or candidate keys
 - D. A candidate key may not be a primary key
17. Assuming the maximum value for the varchar datatype is 255, why would it be necessary to use any value other than 255 in the definition?
- A. Varchar(50) uses much less space than varchar(255)
 - B. You don't think they'd use it so you wouldn't assign the extra space
 - C. The business logic has expectations for the field size
 - D. It wouldn't be necessary – standards say use 255
18. The process of adding a unique value to the beginning or the end of a password before the password is hashed is called _____.
- A. encrypting
 - B. salting
 - C. prepending
 - D. a cipher
19. When using a UNION in a SELECT statement _____.
- A. the two queries on either side of the UNION must be from the same table
 - B. the two queries on either side of the UNION must not be from the same table
 - C. the results are sorted ascending by default
 - D. the results are distinct by default

20. _____ is a data structure technique which allows you to quickly retrieve records from a database file.
- A. Clustering
 - B. Indexing
 - C. Paging
 - D. Transacting
21. Which of the following is NOT a command used to control transactions?
- A. START
 - B. SET TRANSACTION
 - C. COMMIT
 - D. ROLLBACK
22. The acronym _____ representing the standard properties of a transaction?
- A. ACID
 - B. CROSS
 - C. CIDD
 - D. DOTS
23. Which of the answers below is categorized as a data manipulation language command?
- A. CREATE
 - B. SELECT
 - C. INSERT
 - D. COMMIT
24. The SQL statement “_____”, will remove all records from a table and will lock each row prior to the delete.
- A. TRUNCATE TABLE mytable
 - B. DELETE FROM mytable
 - C. ERASE TABLE mytable
 - D. DROP TABLE mytable

Consider the following three tables for questions 25-33

Member's m_id is auto-increment, unique; Item's id is auto-increment, unique; All fields are non-null

member				member_item		item			
m_id	f_name	l_name	status	m_id	item_id	id	descrip	paid	size_cm
1004	Jim	Jones	Active	1117	50	30	Item1	33.50	1102
1117	Jan	Smith	Retired	1551	30	40	Item2	66.25	5
1551	Betty	Shine	Active	1551	55	50	Item3	80.80	5512
				1004	30	55	Item3a	17.71	3232
				1551	75	70	Item5	52.31	16
				1117	30	75	Item5a	100.65	505

25. How many rows would be returned from the following query?

SELECT f_name, l_name, descrip, paid

FROM member a

INNER JOIN member_item b on a.m_id = b.m_id

INNER JOIN item c on b.item_id = c.id

WHERE paid BETWEEN 33.50 and 90;

- A. 5
- B. 4
- C. 3
- D. 2

26. Which query will get a unique list of the items Jan Smith purchased?

A. SELECT DISTINCT descrip

FROM member a

INNER JOIN member_item b ON a.m_id = b.m_id

INNER JOIN Item c ON b.item_id = c.id

WHERE m_id = 1117

B. SELECT DISTINCT (f_name, descrip)

FROM member

INNER JOIN member_item a ON a.m_id = member.m_id

INNER JOIN item ON a.item_id = item.id

WHERE f_name = "Jan"

C. SELECT DISTINCT descrip

FROM member_item a

INNER JOIN member b ON a.m_id = b.m_id

INNER JOIN item c ON a.item_id = c.id

WHERE c.m_id = 1117

D. SELECT DISTINCT descrip

FROM member_item a

INNER JOIN member b ON a.m_id = b.m_id

INNER JOIN item c ON a.item_id = c.id

WHERE f_name = "Jan"

27. The following query produces the record set below and the sum(paid) field is incorrect – why?

```
SELECT status, descrip, sum(paid) AS total
      FROM member a
            INNER JOIN member_item b on a.m_id = b.m_id
            INNER JOIN item c on b.item_id = c.id
      WHERE paid BETWEEN 33.50 and 90
      GROUP BY status, descrip;
```

<u>status</u>	<u>descrip</u>	<u>total</u>
Retired	Item3	80.80000305175781
Active	Item1	67
Retired	Item1	33.5

- A. Using multiple group by columns makes math an approximation
 - B. Floating point math is an approximation
 - C. Using an alias on an aggregate field makes math an approximation
 - D. The number is correct, just shows too many digits
28. How many foreign keys are in the member_item table?
- A. 0
 - B. 1
 - C. 2
 - D. 3

29. The following query produces how many rows?

```
SELECT status, descrip, sum(paid) AS total
      FROM member CROSS JOIN item
```

- A. 3
- B. 9
- C. 18
- D. 36

30. What would “CREATE INDEX master ON member (m_id)” do?
- A. It would add an extra column to the master table with the sort order
 - B. It would create a small lookup table
 - C. It would create an additional sorted version of the master table
 - D. It would sort the master table in place
31. The highest level of normalization met by the member table is _____.
A. 3NF
B. BCNF
C. 4NF
D. 5NF
32. The following query created how many rows?
- ```
SELECT f_name FROM member

UNION

SELECT f_name FROM member
```
- A. 0
  - B. 1
  - C. 3
  - D. 6
33. What would be the result of the following query?
- ```
SELECT descrip  
  
FROM item a  
  
LEFT JOIN member_item b ON a.id = b.item_id  
  
WHERE b.item_id IS NULL
```
- A. No result
 - B. Item2, Item4
 - C. Item3a, Item5a
 - D. Item1, Item2, Item3, Item3a, Item4, Item5, Item5a

34. Which is NOT a benefit of database normalization?
- A. Minimizes data redundancy (duplicate data).
 - B. Removes all null values from your data.
 - C. Results in a more compact database (due to less data redundancy/null values).
 - D. Minimizes / avoids data modification issues
35. NoSQL stands for _____.
- A. Numerical SQL
 - B. No SQL
 - C. Not Only SQL
 - D. Normal SQL
36. A _____ join is formed between two tables where the values of identically named and defined columns are equal.
- A. Natural
 - B. Complete
 - C. Just
 - D. Proper
37. The SQL language has the ability to do conditional statements using _____.
- A. If – Then - Else
 - B. Verify – Else
 - C. Ternary operators
 - D. Case – When - Then – Else
38. Which of the following is NOT true about a temporary table in SQL?
- A. The table gets deleted once the session in which it was created ends
 - B. Temp tables are created with SELECT INTO <tableName>
 - C. Temporary tables can be manually deleted at any time
 - D. You cannot alter a temporary table once its created
39. Constraints can be added to a table with both a CREATE TABLE statement and an ALTER TABLE statement
- A. True
 - B. False
40. Which of the following is a table constraint in SQL?
- A. CREATE
 - B. DEFAULT
 - C. JOIN
 - D. EXISTS

41. What is a DBA when referring to SQL?
- A. Database Administrator
 - B. Data By Additive
 - C. Doing Business Applications
 - D. Driven By Association
42. What is meant by a database trigger?
- A. A log that a specific table was accessed
 - B. A log that a potentially mischievous connection was made to the database
 - C. A statement executed automatically when a table is modified
 - D. A batch process that runs at a specific time and date
43. The situation where a data is stored in the database at a particular moment of time is called a(n) _____.
- A. record
 - B. update
 - C. opportunity
 - D. Instance
44. An advantage of a database view is it _____.
- A. allows for easier database updates
 - B. reduces the dependency on pessimistic locks
 - C. provides consistency to applications
 - D. allows for quicker updates and reads
45. What can be said of the following two queries?
- SELECT fname, MAX(cost) FROM theTable
- SELECT fname, cost FROM theTable ORDER BY cost DESC LIMIT 1
- A. The queries will return the exact same values
 - B. The queries may return the exact same values
 - C. The queries cannot return the exact same value
 - D. The first query will get the max, second will get the min
46. A _____ table in SQL results from turning rows into columns.
- A. Conversion
 - B. Wide
 - C. Pivot
 - D. Transitional

Consider the following tables for questions 47-50

Member's member_id is auto-increment, unique; Item's id is auto-increment, unique;

Empty fields are NULL (not spaces)

member_no	name	dob
123		12/31/1999
456	Jimmy Newtron	01/01/1970
678		11/11/2011
901	Wanda Smith	04/15/1955
998	Tabitha Cramer	03/10/2000

47. What does the following query return?

```
SELECT member_no, name
FROM theTableAbove
WHERE name = NULL
```

- A. 0 rows
- B. 2 row
- C. 5 rows
- D. Syntax error

48. What does the following query return?

```
SELECT dob
FROM theTableAbove
WHERE member_no= '901'
```

- A. 04/15/1945
- B. 11/11/2011
- C. 04/15/1955
- D. 12/31/1999

49. The table already has a primary key on the member_no. How would we set up a composite key on this table?

- A. CREATE COMPOSITE INDEX myComposite ON theTableAbove (name)
- B. CREATE INDEX myComposite ON theTabelAbove (name); CREATE INDEX myComposite ON theTabelAbove (dob)
- C. CREATE INDEX myComposite ON theTabelAbove (member_no)
- D. CREATE INDEX myComposite ON theTabelAbove (name, dob)

50. The theTableAbove meets the highest standardized level of normalization – 5NF.

- A. True
- B. False